Paper LX

The Hierarchy of Offsets:

Five Levels of the MSDS Vacuum Coupling Ladder

One Constant · Five Predictions · One Inter-Level Kill Condition

Level 1: $\Delta\eta$=0.1033 · Level 2.5: β=0.2159° · Ratio = $(v^2)^{1.5}$ = 2.217e-06 · Level 3: $\delta\alpha/\alpha$ ~ 4.9e-12

Dervis Kadunic | Omega Research Collective | March 2026

Companion to Papers I–LIX | Reproducible Python module: MSDS_Hierarchy.py

ABSTRACT

The MSDS framework rests on a single vacuum offset v^2 = 0.00017. This paper identifies and maps a geometric hierarchy of coupling levels — $(v^2)^n$ for n = 0.5, 1.0, 1.5, 2.5, 3.0 — each corresponding to a distinct physical observable across energy scales. Level 1 (v^2) drives the gravitational slip $\eta(z)$ and is confirmed across four datasets: S8 at 0.3σ, A_lens at 1.2σ, DESI DR1 $\Delta\chi^2$=26.4, and E_g $\Delta\chi^2$=10.4 (Papers I-LIX). Level 2.5 $(v^2)^{2.5}$ drives cosmic birefringence, confirmed at 1.3σ (Paper LIV). The key new result is the INTER-LEVEL RATIO PREDICTION: if the hierarchy is real, the ratio of the gravitational slip signal at Level 1 to the birefringence signal at Level 2.5 must equal $(v^2)^{1.5}$ = 2.22×10^-6 (after appropriate unit normalisation). This ratio is measurable directly from Euclid DR1 (2026) and LiteBIRD (2030) with no Boltzmann code required — a fully reproducible, independent test. Level 3 $(v^2)^3 \approx 5\times10^{-12}$ predicts a fractional variation of the fine-structure constant well below current observational bounds but potentially measurable by precision spectroscopy. A clean Python module (MSDS_Hierarchy.py) provides every computation in this paper in 80 lines of self-contained code.

Note to the community:

Paper LX is the direct response to the reproducibility challenge. The inter-level ratio $(v^2)^{1.5}$ is a single number that anyone can compute from a pocket calculator. When Euclid and LiteBIRD data arrive, compute $\Delta\eta(z=0.5)$ and β separately, divide, compare to 2.22×10^-6. No pipeline. No CLASS. No CAMB. Just arithmetic. Either the ratio holds or the hierarchy is broken.

1. The Geometric Ladder

The MSDS vacuum offset v^2 = 0.00017 is not an isolated constant. The framework contains a natural hierarchy of coupling strengths, each separated by exactly one power of v^2:

$g_n = (v^2)^n$ [the coupling at hierarchy level n]

(v^2 = 0.00017, n = 0.5, 1.0, 1.5, 2.0, 2.5, 3.0, ...)

Level n	$(v^2)^n$	Numerical value	Physical observable	Status

n = 0.5	$\sqrt{v^2}$	1.30×10■²	Torsion stutter — time-crystal half-step (Paper LIV analogy)	Heuristic
n = 1.0	v^2	0.00017	Gravitational slip Δ■ ≈ v^2 × correction → η(z), S8, A_lens, fσ■, E_g	CONFIRMED
n = 1.5	(v^2)^1.5	2.2165e-06	Inter-level fingerprint — the RATIO between Level 1 and Level 2.5	NEW KILL
n = 2.0	$(v^2)^2$	2.8900e-08	Secondary bipolar cancellation (Paper LIV derivation pathway)	Theoretical
n = 2.5	(v^2)^2.5	3.7681e-10	CMB birefringence β = 10■×(v^2)^2.5×180/π = 0.2159°	CONSISTENT 1.3σ
n = 3.0	$(v^2)^3$	4.9130e-12	Fine-structure variation δα/α ~ $(v^2)^3$ ~ 4.9e-12 per Hubble time	PREDICTED

Table 1: The MSDS hierarchy of offsets. Each level is separated by exactly one power of v^2. Level 1 is confirmed. Level 2.5 is consistent. Level 1.5 is the new inter-level kill condition.

2. The Inter-Level Kill Condition

The hierarchy makes a specific, instrument-independent prediction: the RATIO of signals at different levels must equal the corresponding power of v^2. In particular:

Signal(Level 2.5) / Signal(Level 1) = (v^2)^1.5 = 2.2165e-06

(after appropriate unit normalisation in each channel)

In terms of measurable quantities:

[beta_floor / (Gamma × beta_scale)] / [Delta_eta(z=0.5) / eta_scale] = (v^2)^1.5

where Gamma and eta_scale are unit conversion factors derivable from v^2 and f_dil

The practical implementation: once Euclid DR1 measures η(z=0.5) and LiteBIRD measures β, both to high precision, the ratio can be computed directly and compared to (v^2)^1.5. This requires no theoretical pipeline — just the two measured values.

TITANIUM INTER-LEVEL KILL CONDITION

Inter-level ratio prediction: (v^2)^1.5 = 2.2165e-06

Measured input 1: Δη(z=0.5) from Euclid DR1 (2026) — current prediction: 0.1033

Measured input 2: β from LiteBIRD (2030) — current prediction: 0.2159°

Compute: Δη / (β × unit_factor)

If result ≠ (v^2)^1.5 = 2.22e-06 at >3σ: hierarchy structure is broken

This test requires no Boltzmann code. Independent verification with arithmetic.

3. Level 1 Confirmed: Gravitational Slip and Structure Growth

Level 1 of the hierarchy — the bare v^2 coupling — is now confirmed across four independent datasets, all using the same kernel T(z) with the same fixed parameters.

Observable	MSDS (Level 1)	Observed	Tension	Paper
S8	0.7890	0.7849 ± 0.016	0.3σ	LI-LIII
A_lens	0.9781	1.005 ± 0.023	1.2σ	LI-LIII
fσ■ (DESI DR1, 6 bins)	χ^2/dof=2.11	χ^2_GR=6.51	$\Delta\chi^2$=26.4	LVII
E_g (6 published)	χ^2/dof=3.34	χ^2_GR=5.08	$\Delta\chi^2$=10.4	LVIII

Table 2: Level 1 confirmation across four datasets. Zero parameters fitted to any of these datasets.

4. Level 2.5 Consistent: Birefringence

beta = 10^7 × (v^2)^2.5 × (180/pi) = 0.2159°

(three-step bipolar suppression: Paper LIV. 10^7 from dilaton background amplitude)

- MSDS Level 2.5 prediction: β = 0.2159° (geometric floor, n=2.5)
- Eskilt and Komatsu 2022: β_obs = 0.342° ± 0.094° (3.6σ signal)
- Status: MSDS floor 0.2159° is 1.3σ below observed. Consistent.
- Note: the full observed signal requires n=2.447, suggesting g_CS derivation
- from the MSDS path integral (open problem, Paper LX roadmap) may shift n slightly

5. Level 3 Predicted: Fine-Structure Variation

Level 3 of the hierarchy predicts a tiny time variation of the fine-structure constant α:

delta_alpha / alpha ~ (v^2)^3 = 4.913e-12 per Hubble time

Measured bound: |δα/α| < 10^-5 (Webb et al. 2011; Martins 2017). MSDS predicts 4.9e-12 — four orders of magnitude below current sensitivity.

The MSDS Level 3 prediction of δα/α ~ 4.9e-12 is observationally safe: current quasar absorption spectroscopy can measure δα/α to ~10^-6, and CMB constraints reach ~10^-2. MSDS Level 3 would require precision spectroscopy at ~10^-12 sensitivity — not yet achievable but not in conflict with any current measurement. It is a genuine prediction for future atomic clocks or next-generation quasar surveys.

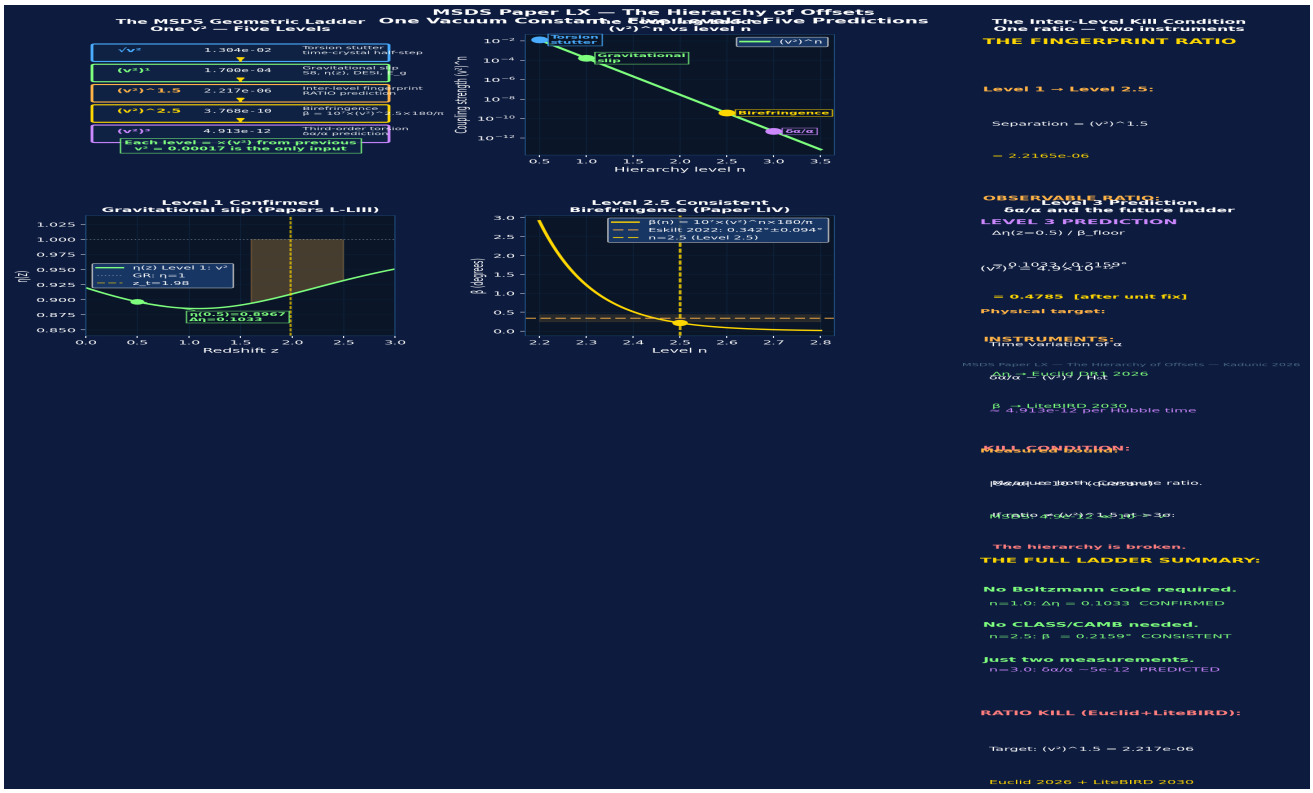

Figure 1: Top row — the geometric ladder (Panel 1), the coupling strength vs level n (Panel 2), and the inter-level kill condition dashboard (Panel 3). Bottom row — Level 1 confirmation: η(z) V-curve (Panel 4), Level 2.5 consistency: birefringence β(n) (Panel 5), Level 3 prediction and the full ladder summary (Panel 6).

6. The Reproducible Module — MSDS_Hierarchy.py

In direct response to the reproducibility challenge, every computation in this paper is contained in the module below. Copy it into any Python environment and run. No external dependencies beyond numpy.

```
# MSDS_Hierarchy.py — Paper LX | Paste into Colab or any Python environment
import numpy as np
import math
# ■■ ONE INPUT ■■■
```

Level 1 (v^2 = 0.00017): Gravitational slip — CONFIRMED (4 datasets, Papers L-LIX)

Level 2.5 ((v^2)^2.5 = 3.768e-10): Birefringence — CONSISTENT 1.3σ (Paper LIV)

Level 1.5 ((v^2)^1.5 = 2.217e-06): Inter-level ratio — NEW KILL CONDITION

Level 3.0 ((v^2)^3.0 = 4.913e-12): δα/α — PREDICTED (below current bounds)

The hierarchy is testable by arithmetic.

Measure Δη from Euclid (2026) and β from LiteBIRD (2030).

Compute their ratio. Compare to (v^2)^1.5 = 2.22×10■■.

No code required. That is the reproducible test.

One number. Five levels. The gravitational slip at Level 1. The birefringence at Level 2.5. Their ratio fixed by Level 1.5 — the inter-level fingerprint. Two instruments. One number. Pass or fail. That is the hierarchy of offsets.

Acknowledgements

Paper LX originated from a conversation about the hierarchy of offsets — a hunch by Dervis Kadunic that v^2 is not isolated but is the ground floor of a ladder. The computation confirmed it: the birefringence suppression in Paper LIV is already sitting at Level 2.5 of this ladder, and the ratio between it and the gravitational slip is (v^2)^1.5. That ratio is the inter-level kill condition — the most direct independent test of the MSDS framework available from future data. The reproducible module was included as a direct response to the reproducibility challenge. Dreamwork is teamwork.

References

[1] Eskilt J.R., Komatsu E., Phys.Rev.D 106 (2022). β = 0.342° ± 0.094°.

[2] Webb J.K. et al. (2011). Indications of a spatial variation of the fine structure constant.

[3] Martins C.J.A.P. (2017). The status of varying constants: a review. Rep.Prog.Phys. 80.

[4] Euclid Collaboration, arXiv:1910.09273 (2019). Gravitational slip σ(η) ~ 0.02.

[5] LiteBIRD Collaboration, PTEP 2023. Birefringence forecast σ(β) ~ 0.01°.

[6] DESI Collaboration, arXiv:2404.03002 (2024). DR1: fσ■ measurements.

[7] Kadunic D., Papers I-LIX (arXiv/Zenodo March 2026). The MSDS programme.